

Zohaib Ali Mughal

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SUMMARY

Full-stack engineer specializing in end-to-end TypeScript (Next.js, NestJS, Fastify, tRPC, Prisma, PostgreSQL) and Flutter. Since late 2024, architected and shipped three production platforms end to end — a venue-booking SaaS with mobile apps live on both app stores, a staffing marketplace running Stripe Connect payouts, and a field-service platform built around a financial-enforcement engine — owning everything from schema design to CI/CD and store release.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Dart, SQL, Python

Frontend: Next.js 16, React 19, Tailwind CSS, TanStack Query, Radix / shadcn UI

Mobile: Flutter 3 (BLoC, clean architecture), Capacitor 6, App Store & Play Store releases

Backend: Node.js, NestJS 11, Fastify 5, tRPC 11, REST / OpenAPI, Zod, BullMQ, NATS JetStream

Data & DevOps: PostgreSQL, Prisma 7, Supabase (Auth / Realtime / Storage), Redis, Docker, AWS EC2, Railway, Vercel, GitHub Actions CI/CD

Payments & Quality: Stripe Connect & Billing, Paymob, Sentry, Jest, Vitest, Playwright, Testcontainers

PROFESSIONAL EXPERIENCE

Slidey LLC | *Lead Full-Stack Software Engineer*

Nov 2024 – Present

Architecting, building, and operating multiple production B2B/B2C SaaS platforms end to end.

ETG Platform — Field-Service Operations & Enforcement

Mar 2026 – Present

Greenfield platform for an Australian electrical/security/AV group (two legal entities sharing customers), built to stop an estimated \$5k/week of margin leakage from unbilled supplier costs and scheduled-vs-actual billing gaps.

- Sole architect of a 12-package Turborepo with 3 deployable apps (Next.js 16 frontend, Fastify 5 API, BullMQ worker) — 31 tRPC routers and a 50+ model Prisma schema spanning jobs, dispatch, service tickets, quoting, and reconciliation.
- Designed an enforcement-first decision engine: one pure, unit-testable gate evaluates every dispatch, closeout, and invoice transition, returns structured failure codes, and writes every block and manager override to an audit trail (8 live checks on the dispatch calendar alone).
- Built scored two-hop reconciliation matching bank transactions to supplier invoices, with “golden-rule” gates so a job cannot reach ready-to-invoice until its costs are accounted for; all money handled as BigInt integer cents (ex-GST).
- Own infrastructure end to end: docker-compose local parity (Postgres, Redis, MinIO), GitHub Actions CI building GHCR images, deploys to Railway and Vercel with automated post-deploy production smoke tests.

Stack: Next.js 16, React 19, Fastify 5, tRPC 11, Prisma 7, PostgreSQL, BullMQ, Redis, Docker.

Hyss — B2B Live-Event Staffing Marketplace (hyss.io)

Jan 2026 – Present

Production SaaS where event companies hire certified freelance crew: competency-matched shifts, call sheets, realtime chat, invoicing, and payouts.

- Spearheading a type-safe Turborepo — 93 Prisma models, ~340 tRPC procedures — serving web, iOS, and Android from a single Next.js codebase via Capacitor, including a custom native Android Credential Manager plugin.
- Built the Stripe Connect financial core: payout onboarding, hybrid seat + usage-metered subscription billing, per-invoice checkout links with three fee-split modes, and idempotent webhook reconciliation.
- Implemented bidirectional Google + Apple (CalDAV) calendar sync — push-channel updates, AES-256-GCM-encrypted OAuth tokens, a retryable row-locked job queue — plus realtime per-event crew chat on Supabase Realtime.
- Hardened security and quality: company-enforceable TOTP MFA with hashed recovery codes, tokenized

document-access approvals, and CI running ephemeral-Postgres integration tests and full-stack Playwright E2E on every PR.

Stack: Next.js 16, React 19, Fastify 5, tRPC 11, Prisma 7, PostgreSQL, Supabase, Stripe Connect, Capacitor 6, Sentry.

Sportefy — Sports Venue Booking & Matchmaking SaaS

Mar 2025 – Dec 2025, now on retainer

B2B2C platform live in Pakistan: NestJS API, Next.js admin dashboard, and a Flutter app on the App Store and Play Store.

- Built the backend to 42 NestJS modules and 100+ versioned REST endpoints with an additive-only /v1 contract, OpenAPI drift gates in CI, and server-driven version gating — every release shipped without breaking the mixed-version mobile install base.
- Developed the consumer Flutter app (21 feature modules, 414 passing tests): offline request queueing with replay, automatic token refresh, fault-tolerant parsing that drops malformed rows instead of crashing screens, QR check-in, and wallet checkout.
- Engineered the payments domain — pluggable gateway adapter (Paymob), wallet/credits ledger, promo codes, PDF receipts — and realtime match chat and notifications via Supabase Realtime with Postgres row-level security, no custom websocket infrastructure.
- Implemented multi-actor RBAC (5 platform roles plus facility/venue-scoped staff roles) from one shared permission matrix consumed by both API guards and the admin dashboard; owned Dockerized AWS EC2 deploys from GitHub Actions.

Stack: NestJS 11, Flutter 3.8, Next.js 16, Prisma 7, PostgreSQL, Supabase, AWS EC2, GitHub Actions.

A-Safe Digital — Event-Driven Microservices Nov 2024 – Feb 2025 *CRM and quoting-engine microservices, previously bottlenecked by monolithic dependencies under high load.*

- Maintained and scaled an event-driven microservices architecture using NATS JetStream, enforcing strict type-safe boundaries and clean-architecture principles.
- Achieved <100ms inter-service latency and authored 500+ unit/integration tests to maintain 95% code coverage, eliminating critical production regressions.

Stack: TypeScript, Node.js, NATS JetStream, React, PostgreSQL.

TECHNICAL PROJECTS

RiskSense — Loan-Default MLOps Pipeline

Final-Year Project, Meezan Bank Limited

- Engineered an ensemble of five ML models orchestrated with ZenML and MLflow (SMOTE balancing, KNN imputation), reaching 92% AUC-ROC with <100ms inference via a containerized Flask API.

Stack: Python, ZenML, MLflow, Docker, PostgreSQL, Flask.

Algorithmic Chess Engine

- Implemented min–max search with alpha–beta pruning, killer-move heuristics, and transposition tables — cutting search branching by 75% and reaching a verified 1600 ELO.

Stack: Python, algorithms & data structures.

ACADEMIC ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA Karachi):** mentored 100+ students and translated complex architecture concepts into hands-on workshops, raising class assessment averages by 30%.
- **Kaggle ML challenges:** secured 2nd and 3rd place finishes across two university-wide machine-learning competitions (150+ participants), rapidly prototyping predictive models under constraints.

EDUCATION

Institute of Business Administration (IBA), Karachi

Graduated: 2025

BSc in Computer Science | National Talent Hunt Program Scholar (Top 33 / 7,290 candidates)